

OPERATING SYSTEMS

Memory Management

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OPERATING SYSTEMS

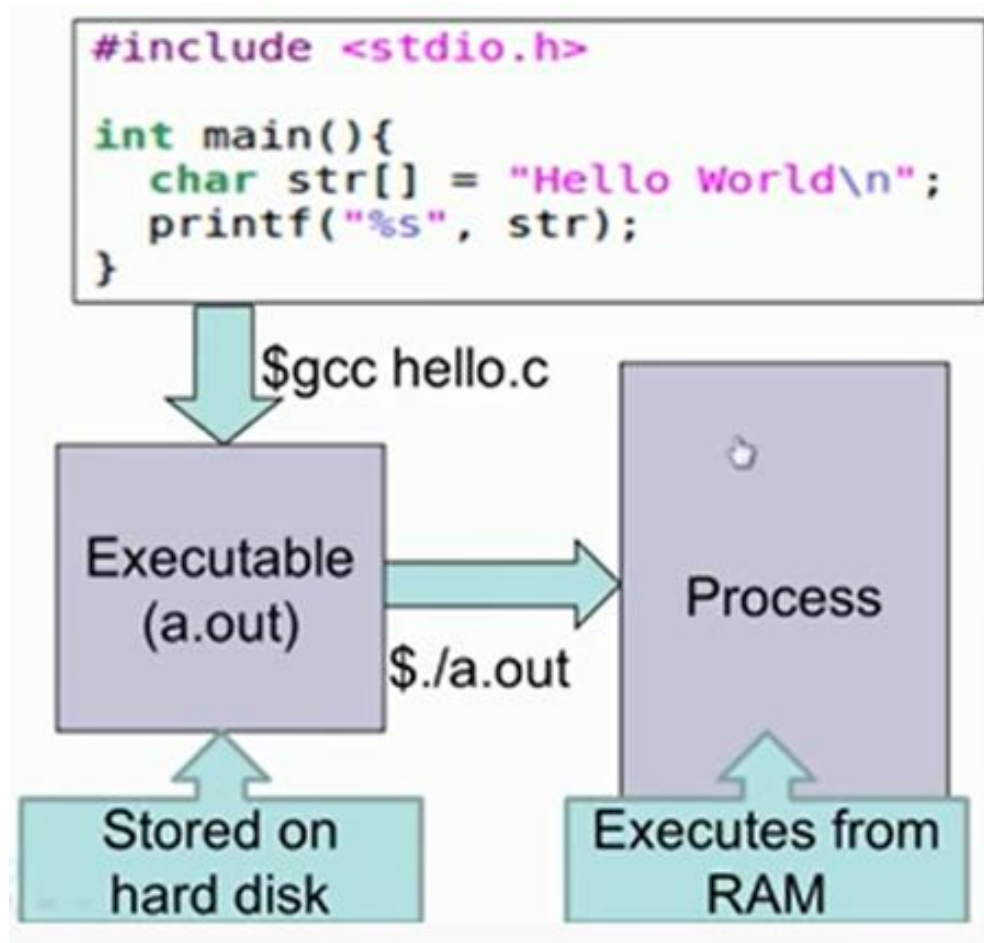
Background

- Basic Hardware
- Address Binding
- Physical Vs Logical Address Space



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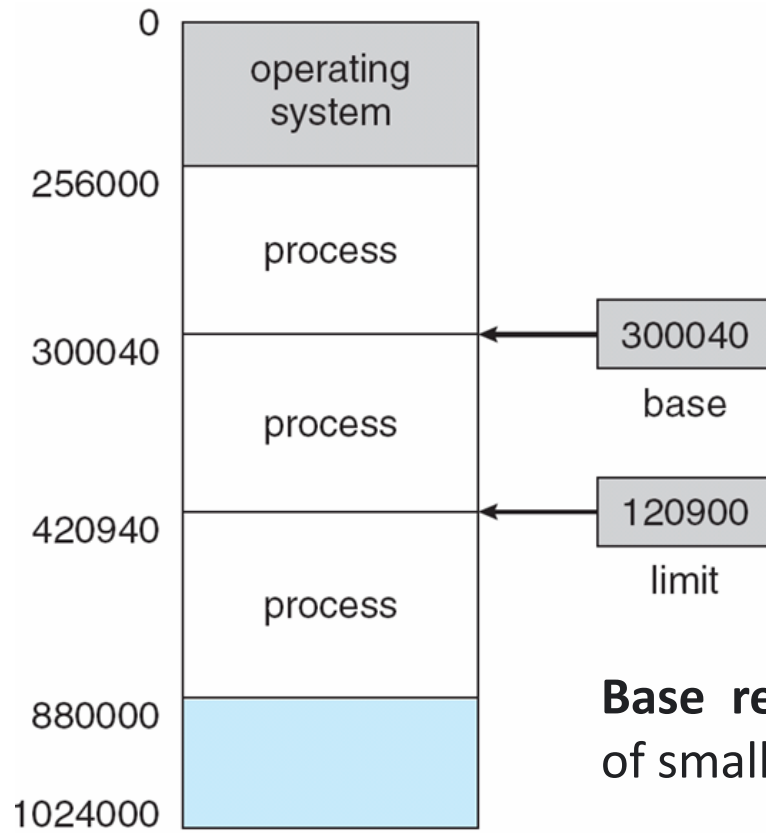
Background



Basic Hardware

- Program must be brought (from disk) into memory and placed within memory for it to be run
- Main memory and registers are only storage that CPU can access directly
- Register access in one CPU clock (or less) whereas Main memory access can take many cycles
- A pair of **base** and **limit** registers define the physical address space

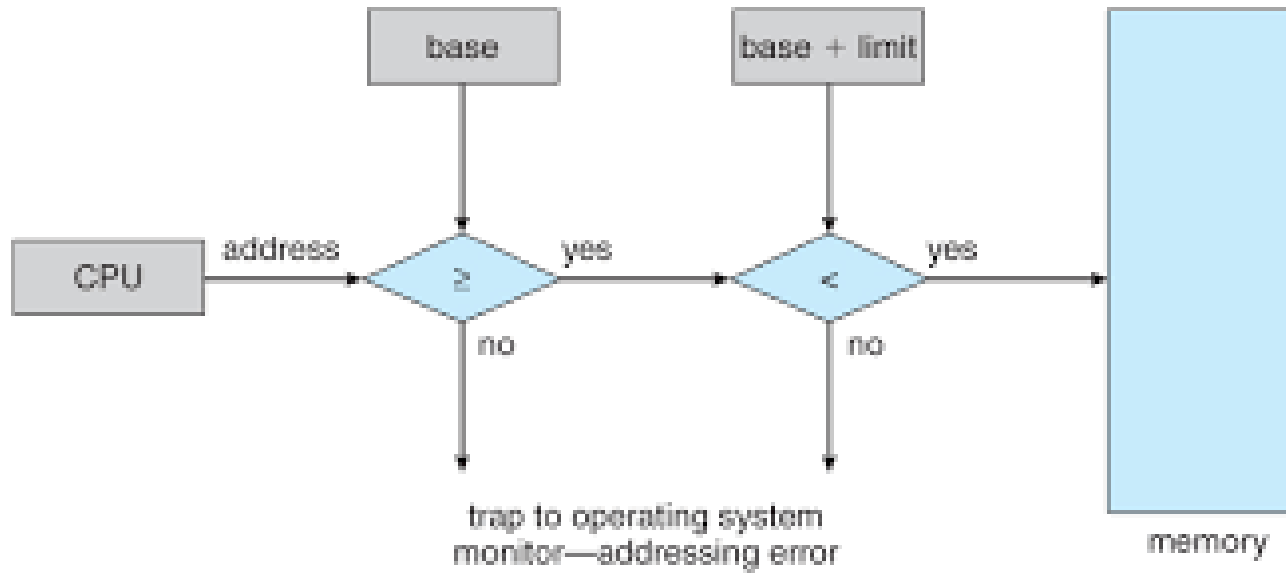
Base and limit registers



Base register contains value of smallest physical address

Limit register contains range of logical addresses

Hardware Support for relocation and limit registers



Address Binding

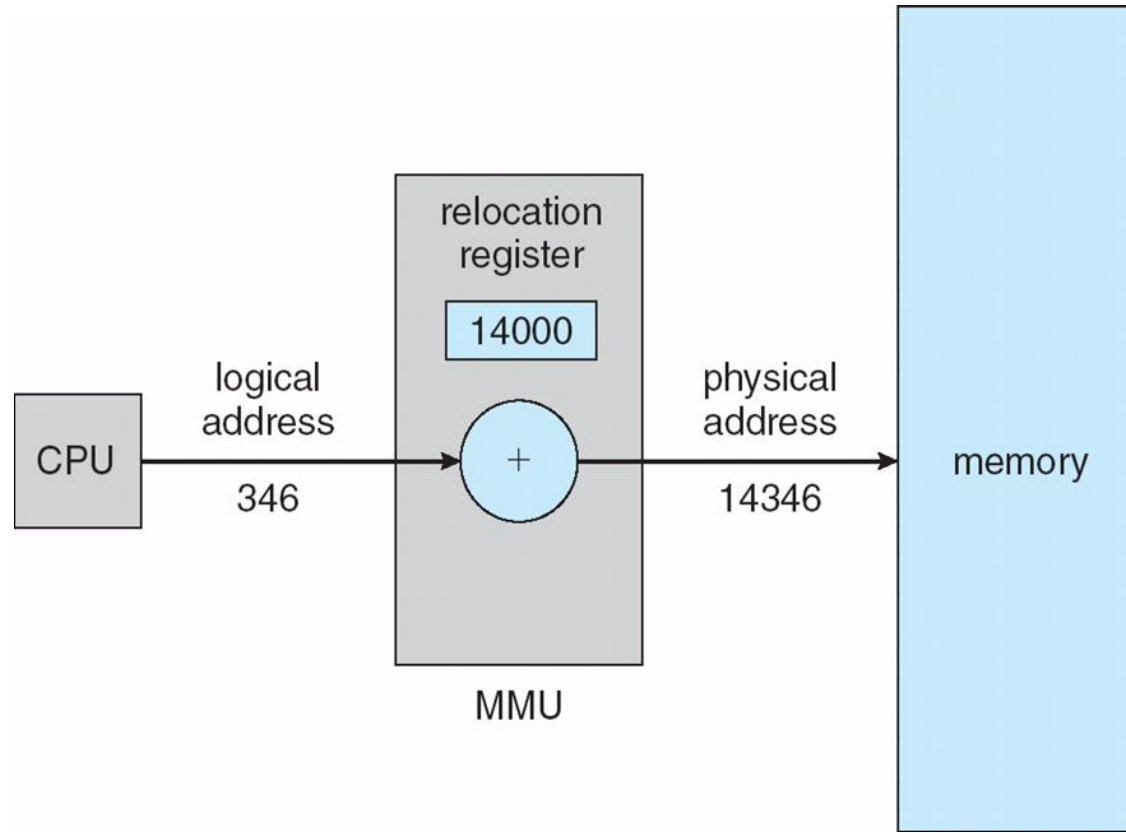
- Address binding of instructions and data to memory addresses can happen at three different stages
 - **Compile time:** If memory location known a priori, **absolute code** can be generated; must recompile code if starting location changes
 - **Load time:** Must generate **relocatable code** if memory location is not known at compile time
 - **Execution time:** **Binding delayed until run time** if the process can be moved during its execution from one memory segment to another. Need hardware support for address maps (e.g., base and limit registers)

Logical Vs Physical Address Space

- **Logical address** – generated by the CPU; also referred to as **virtual address**
- **Physical address** – address seen by the memory unit
- Logical and physical addresses are the same in compile-time and load-time address-binding schemes
- Logical (virtual) and physical addresses differ in execution-time address-binding scheme
- Hardware devices maps virtual and physical addresses
- This hardware will be managed by MMU
- The user program deals with *logical* addresses; They never see the *real* physical addresses

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THANK YOU

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